

Exact zeros of weighted sums of completely multiplicative functions

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Abstract. A completely multiplicative function ε with values ± 1 can make $\sum_{n \leq m} n \varepsilon(n)$ vanish only when $m \equiv 0$ or $3 \pmod{4}$. We prove that for every such m there is an ε_m , depending on m , with this sum zero, that ε_m can be chosen with its signs at finitely many primes prescribed in advance once m is large, and that the functions with infinitely many zeros form a dense G_δ set of probability zero; unweighted sums, by contrast, vanish infinitely often almost surely, by a theorem of Aymone on their sign changes. The proof combines a second moment with arithmetic progressions in subset sums of primes. Quadratic characters, extended completely multiplicatively, give functions with infinitely many zeros. As a consequence, for every prime $q \equiv 7 \pmod{8}$ Conrey's sine series of the Legendre symbol modulo q vanishes identically on an interval.

1 Introduction

How often the partial sums of a completely multiplicative function $\varepsilon : \mathbb{N} \rightarrow \{\pm 1\}$ vanish has a recent answer for random ε . Aymone proved that for a Rademacher random completely multiplicative function the sums $\sum_{n \leq x} \varepsilon(n) n^{-\alpha}$ change sign infinitely often, almost surely, for each fixed $0 \leq \alpha < 1/2$ [Aym24, Thm. 2, p. 896]; for $\alpha = 0$ the steps are ± 1 , so these sums vanish infinitely often. Klurman, Munsch and Sun studied how small the harmonic sums can be [KMS26]. This note is about the opposite weight,

$$S_\varepsilon(m) = \sum_{n \leq m} n \varepsilon(n). \quad (1)$$

A change of sign of S_ε no longer forces a zero, and an exact zero is a solution of a signed subset-sum problem whose signs are tied together by multiplicativity.

The same weighted sums enter a sine series of Conrey. In an approach to the Riemann hypothesis through Legendre symbols, he studied the sine series $f_q(x) = \sum_{n \geq 1} (\frac{n}{q}) \sin(2\pi nx)/n^2$ and showed, for squarefree $q \equiv 3 \pmod{8}$ with $q > 3$, that $f_q(x) = 0$ forces either $\sum_{n \leq qx} (\frac{n}{q}) = h(-q)$ and $\sum_{n \leq qx} n (\frac{n}{q}) = 0$, or a second, isolated condition on qx ; he remarked that the first alternative would make f_q vanish identically on an interval [Con24, Thm. 4, Cor. 1 and end of §6]. For $q \equiv 3 \pmod{8}$, the case of his question, we do not decide whether this happens: it is excluded under a positivity hypothesis of the kind he proposes (Proposition 4.6) and is otherwise open (Question 6.1). His formula applies equally to primes $q \equiv 7 \pmod{8}$, and there the answer is a statement about exact zeros of the weighted sums of one fixed completely multiplicative function: f_q vanishes on an interval for every such q (Corollary C).

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Changing one sign changes $S_\varepsilon(m)$ by an even amount, so $S_\varepsilon(m) \equiv m(m+1)/2 \pmod{2}$, and a zero is possible only when $m \equiv 0$ or $3 \pmod{4}$; call these lengths *admissible*.

Theorem A *For every $m \geq 3$ with $m \equiv 0$ or $3 \pmod{4}$ there is a completely multiplicative $\varepsilon : \mathbb{N} \rightarrow \{\pm 1\}$ with*

$$\sum_{n \leq m} n \varepsilon(n) = 0.$$

The function depends on m : no single ε vanishes at every admissible length, since $S_\varepsilon(4) - S_\varepsilon(3) = 4\varepsilon(4) = 4$. Prescribed signs are no obstruction: for every finite set of prescribed signs at the primes and every large admissible m there is an ε_m with those signs and $S_{\varepsilon_m}(m) = 0$ (Proposition 3.1). For a fixed function, zeros are rare in measure and generic in category.

Theorem B *On $\Omega = \{\pm 1\}^{\mathcal{P}}$, the sign patterns on the primes, let $\mathcal{Z}_\infty$ be the set of ε with $S_\varepsilon(m) = 0$ for infinitely many m . With the product topology, $\mathcal{Z}_\infty$ is a dense G_δ . With independent fair signs, $\Pr(\mathcal{Z}_\infty) = 0$.*

Quadratic characters give explicit points of $\mathcal{Z}_\infty$. For an odd prime q and $s = \pm 1$ let $\chi_{q,s}$ be the completely multiplicative function equal to the Legendre symbol $(\cdot \mid q)$ on the integers prime to q , with $\chi_{q,s}(q) = s$.

Corollary C *Let $q \equiv 7 \pmod{8}$ be prime and $\chi = \chi_{q,s}$ with $s = \pm 1$. Then $S_\chi\left(\frac{q^j-1}{2}\right) = 0$ for every $j \geq 1$, and Conrey's function $f_q(x) = \sum_{n \geq 1} \left(\frac{n}{q}\right) \sin(2\pi nx)/n^2$ vanishes identically on the interval $\left[\frac{q-1}{2q}, \frac{q+1}{2q}\right]$.*

The case $j = 1$ of Corollary C is classical: it is the equality of the sums of the quadratic residues and non-residues below $q/2$ for $q \equiv 7 \pmod{8}$ [AC17, Thm. 1(a)]. The iteration in j and the null interval of f_q are the content of the corollary. It comes from a digit recursion in base q (Theorem 4.1) and from the Euler factor at 2 in the weighted half sum of an odd character (Theorem 4.2). The recursion gives further infinite families of zeros: for $q \equiv 1 \pmod{4}$, for $q \equiv 3 \pmod{4}$ of class number one, along longer loops of digits (at $q = 59, 131, 347$, of class numbers 3, 5, 5), and for $s = -1$ at every $q \equiv 3 \pmod{8}$, at the lengths $(q^{2^i} - 1)/2$ (Corollary 4.3, Proposition 4.4, Table 2). In Conrey's case $q \equiv 3 \pmod{8}$ the null intervals are excluded under a positivity hypothesis of the kind he proposes (Proposition 4.6). Section 5 shows that every admissible length is the length of a vanishing weighted sum of Legendre symbols, for infinitely many primes.

Relation to previous work.

Aymone's theorem [Aym24] gives zeros of the unweighted sums infinitely often, almost surely; Theorems A and B show that with the weight n the picture is reversed: every admissible length is a zero of some function, depending on the length, but a fixed function vanishes infinitely often only on a set of probability zero. For the unweighted

sums, Borwein, Choi and Coons read the partial sums of $\chi_{q,1}$ off the base- q digits [BCC10, Thm. 8 and Cor. 3], and Klurman, Mangerel, Pohoata and Teräväinen give the recursion for any value at q [KMPT21, proof of Lemma 2.8]; the unweighted partial sums of Legendre symbols are also studied as random-like paths [HL24] and for their sign at fixed fractions of q [Kal19]. The weight adds a term to the recursion, and with it the zeros. Theorem 4.2, the weighted half sum, is in Kanemitsu and Kuzumaki [KK01] and in Kučera [Kuč01], and the equality of the sums of quadratic residues and non-residues behind the case $j = 1$ of Corollary C goes back to Lebesgue [Leb42, p. 144, (16)]; see also [Ber76, Cor. 13.4]. The proof of Theorem A uses Sárközy's progressions in subset sums [Sár94], with Lev's constants [Lev03] in the form of Bringmann and Wellnitz [BW21, Thm. 4.23], and that of Theorem B the anticoncentration theorem of Sárközy and Szemerédi [SS65]. The sums $\sum_{c \leq m} c \chi(c)$ arise as the components of a moment matrix attached to rings of character values [Mar26].

Section 2 proves Theorem A, Section 3 Theorem B, and Section 4 Corollary C, Section 5 treats the Legendre symbols at a fixed length, and Section 6 states two questions.

2 Every admissible length

Throughout, ε is a completely multiplicative function $\mathbb{N} \rightarrow \{\pm 1\}$, determined by its values at the primes, and $S_\varepsilon(m)$ is the sum (1).

2.1 The large primes enter once

Everything rests on one observation. A prime $q \in (m/2, m]$ divides no $n \leq m$ other than q itself, so its sign enters $S_\varepsilon(m)$ once, with weight q . For a set Q of such primes,

$$S_\varepsilon(m) = R_Q + \sum_{q \in Q} q \varepsilon(q), \quad (2)$$

where R_Q depends only on the signs outside Q . Since $S_\varepsilon(m)$ is even at admissible lengths, $R_Q \equiv P_Q \pmod{2}$, with $P_Q = \sum_{q \in Q} q$. The signed sums $\sum \pm q$ are the numbers $P_Q - 2x$, x a subset sum of Q . So the proof has two halves: make $|R_Q|$ small by the signs outside Q , and show that the subset sums of Q contain a long interval $[\alpha, P_Q - \alpha]$; then every integer of the parity of P_Q in $[-D, D]$, $D = P_Q - 2\alpha$, is a signed sum, and $|R_Q| \leq D$ gives a zero.

Lemma 2.1 *Give the primes outside Q independent uniform signs. Then*

$$\mathbb{E} R_Q^2 \leq \frac{m^3 (\log m + 13)}{9},$$

so some choice has $|R_Q| \leq B_*(m) := \frac{1}{3} m^{3/2} \sqrt{\log m + 13}$.

Proof. $\mathbb{E} \varepsilon(n) \varepsilon(k)$ is 1 if nk is a square and 0 otherwise. Writing $n = da^2$ with d squarefree, $\mathbb{E} S_\varepsilon(m)^2 = \sum_d b_m(d)^2$ with $b_m(d) = d \sum_{j \leq \sqrt{m/d}} j^2$, and $\mathbb{E} R_Q^2$ is the same sum without the terms $d = q \in Q$, whose class contains only $n = q$. With

$J = \lfloor \sqrt{m/d} \rfloor$, $\sum_{j \leq J} j^2 \leq J^3/3 + J^2/2 + J/6$, and $dJ^2 \leq m$, $dJ \leq \sqrt{md} \leq m$ give $b_m(d) \leq m^{3/2}/(3\sqrt{d}) + 2m/3$. Squaring and summing over $d \leq m$ with $\sum 1/d \leq 1 + \log m$, $\sum d^{-1/2} \leq 2\sqrt{m}$ and at most m terms gives $m^3((1 + \log m) + 8 + 4)/9$. ■

Lemma 2.2 Write $\Sigma(A)$ for the set of subset sums of a finite set A of positive integers.

- (a) If $\Sigma(A) \supseteq [\alpha, \beta]$ and $x \notin A$, $0 < x \leq \beta - \alpha + 1$, then $\Sigma(A \cup \{x\}) \supseteq [\alpha, \beta + x]$. As $\Sigma(A)$ is symmetric under $x \mapsto P_A - x$, $\Sigma(A) \supseteq [\alpha, \beta]$ with $\beta \geq P_A/2$ implies $\Sigma(A) \supseteq [\alpha, P_A - \alpha]$.
- (b) (pairs) Let $q_0, u_1, v_1, \dots, u_t, v_t$ be distinct positive odd integers, $d_j = (v_j - u_j)/2$ with $1 \leq d_j \leq 1 + d_1 + \dots + d_{j-1}$, $H = \sum d_j$, $U = \sum u_j$, and $q_0 \leq 2H + 1$. Then $\Sigma(\{q_0, u_j, v_j\}) \supseteq [U + q_0 - 1, U + 2H + 1]$.
- (c) (Sárközy, with Lev's constants [BW21, Thm. 4.23]) If X is a set of n positive integers with maximum x^* and $4x^* \leq \ell \leq n^2/(12 \log_2(4x^*/n))$, then $\Sigma(X)$ contains an arithmetic progression of ℓ terms and step $s \leq 4x^*/n$, each term a sum of at most $6\ell/n$ distinct elements of X .

Proof. (a) is immediate. (b) The condition on the d_j makes their subset sums exactly $\{0, \dots, H\}$; choosing one element of each pair gives $U + 2\{0, \dots, H\}$, and adding q_0 or not fills the other parity: an integer x of the interval with $x - U$ even is $U + 2k$, $k \leq H$, and one with $x - U$ odd is $U + q_0 + 2k$ with $0 \leq k \leq H$. (c) is the cited theorem: as $\log_2 y \geq \log y$ for $y \geq 1$, its hypothesis with $\log_2$ implies the hypothesis with the natural logarithm, so (c) holds whichever logarithm the source intends. ■

2.2 The proof

Proof of Theorem A. Table 1 summarises the three ranges.

Up to 20000. For each admissible $m \leq 20000$ a function ε with $S_\varepsilon(m) = 0$ is exhibited and the sum is recomputed exactly: for all but four lengths by starting from $\varepsilon(2) = \varepsilon(3) = -1$ (with at most two signs at primes ≤ 7 changed), balancing the primes of $(\sqrt{m}, m/2]$ and solving (2) exactly with the primes of $(m/2, m]$; the lengths 12, 15, 19, 23 by exhaustive search.

From 5000 to 10^8 . Cut $[5000, 10^8)$ into 104 blocks $[a_j, b_j]$,

$$a_1 = 5000, \quad b_j = \min(\lfloor 11a_j/10 \rfloor, 10^8 - 1), \quad a_{j+1} = b_j + 1,$$

and let Q be the primes of $(b_j/2, a_j]$, which lie in $(m/2, m]$ for every m in the block; the other primes of $(m/2, m]$ are among the random signs of Lemma 2.1. Each block carries a certificate: a prime $q_0 \in Q$ and disjoint pairs in Q as in Lemma 2.2(b), with $2H - q_0 + 2 \geq b$. Every other prime of Q is at most $a \leq b$, so Lemma 2.2(a) extends the interval to $[\alpha, P_Q - \alpha]$ with $\alpha = U + q_0 - 1$. The block is proved for all its m once $D = P_Q - 2\alpha$ exceeds $B_*(b) \geq B_*(m)$, which follows from the integer inequality $90D^2 > b^3(130 + 7e)$ with $e = \lceil \log_2 b \rceil$ (as $\log b < 7e/10$). All 104 certificates pass; the bound $b^3(130 + 7e)/(90D^2)$ on $B_*(b)^2/D^2$ is at most 0.445, reached at the first block.

From 10^8 on. Let $M \geq 10^8$, $L = \log M$, Q the N primes of $(M/2, M]$. By Dusart's bounds $x/(\log x - 1) \leq \pi(x)$ for $x \geq 5393$ and $\pi(x) \leq x/(\log x - 1.1)$ for $x \geq 60184$

[Dus10, Thm. 6.9, (6.6)], $N \geq M/(2L)$ and $P_Q \geq NM/2 \geq M^2/(4L)$. Let $X \subset Q$ have $n = \lfloor N/8 \rfloor \geq M/(17L)$ elements and take $\ell = 4M$: the hypothesis of Lemma 2.2(c) follows from $n^2 \geq 48M \log_2(4M/n)$, which holds once $M \geq 13872L^2 \log_2(68L)$, that is from $M \approx 4.4 \cdot 10^7$. This gives a progression $a + js$, $0 \leq j < \ell$, of step $s \leq 4M/n \leq 68L$ whose terms are sums of distinct elements of X , so $a \leq 24M^2/n$. Add a set Y of $s-1$ further primes of Q . (i) As $s \leq 68L < M/2$, every prime of Q is a unit modulo s . (ii) The subset sums of $s-1$ units meet every class modulo s : adding a unit u to a set $A \ni 0$ of classes gives $A \cup (A+u)$, which is strictly larger unless $A+u = A$, and then A is invariant under the generator u of $\mathbb{Z}/s\mathbb{Z}$, so $A = \mathbb{Z}/s\mathbb{Z}$; starting from $\{0\}$, $s-1$ units reach all s classes. (iii) So for each class c there is $y_c \in \Sigma(Y)$, $y_c \equiv c$, with $0 \leq y_c \leq B := (s-1)M$; an integer $x \in [a+B, a+(\ell-1)s]$ is y_c plus the term $x-y_c$ of the progression, with $c \equiv x-a$, and $\Sigma(X \cup Y)$ contains the whole interval $[\alpha, \alpha+w] = [a+B, a+(\ell-1)s]$, with $\alpha \leq 28M^2/n$ and $w = (\ell-1)s - (s-1)M \geq M$. The remaining primes, each at most M , extend it by Lemma 2.2(a); they are $N - n - s + 1 \geq \frac{7}{8}N - s$ primes above $M/2$, while $X \cup Y$ weighs at most $(\frac{1}{8}N + s)M$. (iv) As $N \geq M/(2L)$ and $s \leq 68L$, $5N > 24s$, which is $(\frac{7}{8}N - s)\frac{M}{2} > (\frac{1}{8}N + s)M$: the remaining primes carry more than half of P_Q , and the symmetry closes the interval to $[\alpha, P_Q - \alpha]$. With $K = 2\alpha \leq 56M^2/n$ and Lemma 2.1,

$$\frac{B_*(M) + K}{P_Q} \leq \frac{4}{3} \frac{L\sqrt{L+13}}{\sqrt{M}} + \frac{3808L^2}{M} < 0.027 \quad (M \geq 10^8),$$

the right side being 0.02669 at $M = 10^8$ and both terms decreasing in M . So $|R_Q| < P_Q - K$ and a zero exists. ■

range of m	method	certificate	margin
$3 \leq m \leq 20000$	explicit ε	10000 lengths	exact
$5000 \leq m < 10^8$	Lem. 2.1, 2.2(a,b)	104 blocks	$B_*^2/D^2 \leq 0.445$
$m \geq 10^8$	Lem. 2.1, 2.2(c)	none	$(B_* + K)/P_Q < 0.027$

Table 1: The proof of Theorem A, range by range. The ranges overlap. The 104 certificates (2128 pairs) and the 10000 explicit functions are generated by programs in the supplementary material and stored there, and a separate program that only reads the stored files checks them in integer arithmetic.

The seam at 10^8 has a reason. The pairs of Lemma 2.2(b) must start with $d_1 = 1$, a pair of twin primes inside $(b/2, a]$: every block has one, but no theorem says every interval will, so the certificates alone cannot cover all m . Sárközy's progression needs no such pair, and it is available from about $4.4 \cdot 10^7$.

3 Along one function

Theorem A lets ε depend on m . Fix it instead, and ask how often $S_\varepsilon(m)$ vanishes.

Proposition 3.1 *Let F be a finite set of primes and $\varepsilon_0 : F \rightarrow \{\pm 1\}$. There is $M(F)$ such that for every admissible $m \geq M(F)$ some ε with $\varepsilon|_F = \varepsilon_0$ has $S_\varepsilon(m) = 0$. For $F = \{2, 3, 5, 7\}$ one may take $M(F) = 60000$.*

Proof. For $m > 2 \max F$ the set F is disjoint from Q . Conditioning on the event $\varepsilon|_F = \varepsilon_0$, of probability $2^{-|F|}$, multiplies $\mathbb{E} R_Q^2$ by at most $2^{|F|}$, since $R_Q^2 \geq 0$; the proof of Theorem A then goes through with $2^{|F|/2} B_*$ in place of B_* , and both B_*/P_Q and K/P_Q tend to 0. For $F = \{2, 3, 5, 7\}$ the factor is 4: every block with $b \geq 60000$ satisfies $90D^2 > 16b^3(130 + 7e)$, and beyond 10^8 , $4B_*/P_Q + K/P_Q < 0.07$. ■

Proof of Theorem B. The set U_M of ε with a zero at some admissible $m \geq M$ is open, each condition involving finitely many signs, and dense by Proposition 3.1; $\mathcal{Z}_\infty = \bigcap_M U_M$ is a dense G_δ by the Baire category theorem, the space of all ε being compact and metrizable. For the measure, condition on every sign outside the primes Q of $(m/2, m]$: by (2), $S_\varepsilon(m) = R_Q + \sum_{q \in Q} q \varepsilon(q)$ with distinct weights, and the theorem of Sárközy and Szemerédi [SS65], [Ngu12, Thm. 1.1] bounds the probability of any single value by $O(|Q|^{-3/2}) = O((\log m/m)^{3/2})$. The series converges, and the first Borel–Cantelli lemma gives $\Pr(\mathcal{Z}_\infty) = 0$. ■

The weight is what makes this true. Without it the steps of $\sum_{n \leq m} \varepsilon(n)$ are ± 1 , a change of sign forces a zero, and by Aymone’s theorem [Aym24, Thm. 2, p. 896] there are almost surely infinitely many; with the logarithmic weight $1/n$ the question becomes one of small values [KMS26]. With the weight n the sum jumps over 0.

4 Quadratic characters

Theorem B says that a function with infinitely many zeros is exceptional in measure. Many natural functions are among the exceptions. For an odd prime q and $s = \pm 1$ let $\chi = \chi_{q,s}$ be as in Section 1, and put

$$C(r) = \sum_{r' \leq r} \chi(r'), \quad T(r) = \sum_{r' \leq r} r' \chi(r') \quad (0 \leq r < q), \quad T_q = T(q-1).$$

4.1 A digit recursion

Theorem 4.1 *For $k \geq 0$ and $0 \leq r < q$,*

$$S_\chi(qk + r) = s q S_\chi(k) + k(T_q + q C(r)) + T(r).$$

Proof. The $n \leq qk + r$ prime to q are $jq + r'$ with $j < k$, $1 \leq r' < q$, and $qk + r'$ with $r' \leq r$; as $\chi(jq + r') = \chi(r')$ and $C(q-1) = 0$, they contribute $kT_q + kq C(r) + T(r)$. The multiples jq , $j \leq k$, contribute $\sum_j jq s \chi(j) = sq S_\chi(k)$. ■

Theorem 4.1, the propagation of zeros along digits in R_q (the first statement of Corollary 4.3 below) and Lemma 2.2(b) are also formalized in Lean 4 with Mathlib, for every completely multiplicative $\chi : \mathbb{N} \rightarrow \mathbb{Z}$ that is periodic modulo q on the residues $1, \dots, q-1$ and has $\sum_{r=1}^{q-1} \chi(r) = 0$; the file is in the supplementary material.

One ingredient is classical. For an odd character the weighted half sum has a closed expression through $B_{1,\chi}$ and the Euler factor at 2; in the form of a sum against the shifted sawtooth it is in Kanemitsu and Kuzumaki [KK01, proof of Thm. 1] and in Kučera [Kuč01, Ex. 2]. We state it for the half sum, with a short proof, because the vanishing criterion is what the zeros need.

Theorem 4.2 (Kanemitsu–Kuzumaki, Kučera) *Let f be odd, let χ be an odd Dirichlet character modulo f , and put $B_{1,\chi} = f^{-1} \sum_{a=1}^f a \chi(a)$. Then*

$$\sum_{c=1}^{(f-1)/2} c \chi(c) = \frac{f}{2} B_{1,\chi} (\chi(2)^{-1} - 1).$$

If moreover χ is primitive, so that $B_{1,\chi} \neq 0$, the sum vanishes if and only if $\chi(2) = 1$.

Proof. Write $T = \sum_{c < f/2} c \chi(c)$, $U = \sum_{c < f/2} \chi(c)$, $W = \sum_{c \text{ even}} \chi(c)$ and $X = \sum_{c \text{ even}} c \chi(c)$, all over $1 \leq c < f$. Pairing $b \mapsto f - b$ exchanges odd and even residues, since f is odd, and $\chi(f - b) = -\chi(b)$; hence $\sum_{b \text{ odd}} b \chi(b) = -fW + X$, and adding X gives $fB_{1,\chi} = 2X - fW$, that is $X = \frac{f}{2}(B_{1,\chi} + W)$. Substituting $c = 2c'$ in the even sums, $W = \chi(2)U$ and $X = \chi(2) \cdot 2T$. Finally $\sum_b \chi(b) = 0$ splits as $\sum_{\text{odd}} + \sum_{\text{even}}$, and the first equals $(2\chi(2) - 1)B_{1,\chi}$ by the same substitution applied to $\chi(2)fB_{1,\chi} = \sum_a a \chi(2a)$; so $-\chi(2)U = (2\chi(2) - 1)B_{1,\chi}$, i.e. $U = (\chi(2)^{-1} - 2)B_{1,\chi}$. Putting the three together, $2\chi(2)T = \frac{f}{2}(B_{1,\chi} + \chi(2)U)$ gives the displayed formula. For χ primitive and odd, $B_{1,\chi} = -L(0, \chi) \neq 0$. ■

For a prime $p \equiv 3 \pmod{4}$, $p > 3$, the unweighted half sum is the class number formula of Jacobi and Dirichlet [Jac32], [Dir38, Werke I, p. 365],

$$h(-p) = \frac{1}{2 - \left(\frac{2}{p}\right)} \sum_{0 < n < p/2} \left(\frac{n}{p}\right), \quad (3)$$

with the same Euler factor at 2.

4.2 Zeros from digits

Corollary 4.3 *Let $R_q = \{0 \leq r < q : T_q + q C(r) = 0 = T(r)\}$. Every $m \geq 1$ whose base- q digits all lie in R_q has $S_\chi(m) = 0$, for both values of s .*

- (a) *If $q \equiv 1 \pmod{4}$, then $0, q - 1 \in R_q$, and S_χ has at least $2^J - 1$ zeros below q^J , hence at least $\frac{1}{2} X^{\log 2 / \log q} - 1$ below X .*
- (b) *If $q \equiv 7 \pmod{8}$, then $(q - 1)/2 \in R_q$, and $S_\chi\left(\frac{q^j - 1}{2}\right) = 0$ for every $j \geq 1$.*
- (c) *If $q \equiv 3 \pmod{4}$, $q > 3$, $h(-q) = 1$ and $s = 1$, then $S_\chi(z) = 0$ along $z_0 = q$, $z_{i+1} = q(z_i + 1)$. For $q = 3$ and $s = 1$ the same holds with 27 in place of q : $z_0 = 27$, $z_{i+1} = 27(z_i + 1)$.*
- (d) *For every q and s : if $S_\chi(qk) = 0$ and $C(r) = T(r) = 0$, then $S_\chi(qk + r) = 0$. When $q > 3$ and $\chi(2) = \chi(3) = -1$ this applies with $r = 4$, since $1 - 2 - 3 + 4 = 0$. For $q = 3$ and $s = -1$, where also $\chi(2) = \chi(3) = -1$, the conclusion holds as well: $S_\chi(3k) = 0$ implies $S_\chi(3k + 4) = 0$.*

- (e) If $s = -1$, then $S_\chi(q^2 m) = q^2 S_\chi(m)$ for every m : a single zero of S_χ gives infinitely many.
- (f) If $q \equiv 3 \pmod{8}$, then $S_\chi\left(\frac{q^j - 1}{2}\right) = h(-q) q^j (1 + s + \dots + s^{j-1})$ for every $j \geq 1$, with $h(-3)$ read as the weighted class number $1/3$. In particular, for $s = -1$, $S_\chi\left(\frac{q^{2i} - 1}{2}\right) = 0$ for every $i \geq 1$.

Proof. By induction on the number of digits, from $S_\chi(0) = 0$ and Theorem 4.1. (a) χ is even, so $r \leftrightarrow q - r$ gives $T_q = 0$; with $C(q - 1) = 0$, both 0 and $q - 1$ lie in R_q . (b) χ is odd and $\chi(2) = 1$. Theorem 4.2 gives $T((q - 1)/2) = 0$, the Euler factor $\chi(2)^{-1} - 1$ vanishing; the half sum (3) gives $C((q - 1)/2) = (2 - \chi(2)) h(-q) = h(-q)$; and $T_q = q B_{1,\chi} = -q h(-q)$. So $T_q + qC = 0$, and $(q^j - 1)/2$ is the number whose j digits are all $(q - 1)/2$. (c) With $r = 0$ and $k = z + 1$, Theorem 4.1 gives $S_\chi(q(z + 1)) = q(z + 1)(s\chi(z + 1) - h(-q))$ when $S_\chi(z) = 0$; $S_\chi(q) = T_q + qs = q(s - h) = 0$, and $z_i \equiv 0 \pmod{q}$ gives $\chi(z_i + 1) = 1$. At $q = 3$, $T_3 = -1$, and three steps with $r = 0$ give $S_\chi(27k) = 27S_\chi(k) - 27k$; so $S_\chi(27) = 0$, and $S_\chi(z) = 0$ with $27 \mid z$ gives $S_\chi(27(z + 1)) = 27(z + 1)\chi(z + 1) - 27(z + 1) = 0$. (d) Subtracting the cases r and 0 of Theorem 4.1, $S_\chi(qk + r) = S_\chi(qk) + kq C(r) + T(r)$; for $q > 3$ the digit $r = 4$ exists. For $q = 3$, $s = -1$ it does not, and directly: $T_3 = 1 - 2 = -1$, so $S_\chi(3k) = -3S_\chi(k) - k$ and $S_\chi(3k) = 0$ forces $3 \mid k$; and, as $\chi(3k + 1) = \chi(3k + 4) = 1$, $\chi(3k + 2) = -1$ and $\chi(3k + 3) = -\chi(k + 1)$, $S_\chi(3k + 4) - S_\chi(3k) = (3k + 3)(1 - \chi(k + 1))$, which vanishes because $k + 1 \equiv 1 \pmod{3}$ gives $\chi(k + 1) = 1$. (e) With $r = 0$ and $s = -1$, $S_\chi(qm) = -qS_\chi(m) + mT_q$; applied twice, the terms in T_q cancel. (f) Put $c = (q - 1)/2$ and $m_j = (q^j - 1)/2$, so that $m_{j+1} = qm_j + c$. Now $\chi(2) = -1$, so Theorem 4.2 gives $T(c) = q h(-q)$, (3) gives $C(c) = 3h(-q)$, and $T_q = -q h(-q)$; at $q = 3$, where $c = 1$, directly $T(1) = C(1) = 1$ and $T_3 = -1$, the same values with $h(-3) = 1/3$. Theorem 4.1 with $k = m_j$ and $r = c$ gives $S_\chi(m_{j+1}) = s q S_\chi(m_j) + q^{j+1} h(-q)$, and $S_\chi(m_1) = T(c) = q h(-q)$. ■

q	R_q	infinite family	first members of the family
5	$\{0, 4\}$	(a), even character	4, 20, 24, 100, ...
13	$\{0, 12\}$	(a)	12, 156, 168, ...
29	$\{0, 4, 24, 28\}$	(a), four digits	4, 24, 28, 116, ...
7	$\{3\}$	(b) and (c)	3, 24, 171, ...; 7, 56, 399, ...
23	$\{11\}$	(b), $h = 3$	11, 264, 6083, 139920, ...
11, 19, 43, 67, 163	$\emptyset$	(c), $h = 1$	11, 132, 1463, 16104, ...
59	$\emptyset$	block, $h = 3$	16708800, ...

Table 2: Infinite families of zeros of S_χ , $s = +1$, given by Corollary 4.3 and Proposition 4.4; for $q = 11, 19, \dots$ the zeros listed are those of $q = 11$. The last column lists members of each family, not all zeros of S_χ ; at $q = 59$ the first zero is $4\,591\,439 = (22, 21, 0, 0)_{59}$, outside the block family. For $q \equiv 7 \pmod{8}$ it is the Euler factor at 2 that puts $(q - 1)/2$ in R_q .

When R_q is empty the recursion can still close a loop on a longer word of digits.

Proposition 4.4 Let $q \equiv 3 \pmod{4}$, $q > 3$, $s = 1$, $h = h(-q)$ and $D(r) = C(r) - h$. Suppose that B has base- q digits $d_{L-1}, \dots, d_0$ with

$$\sum_{i < L} D(d_i) = 0 \quad \text{and} \quad S_\chi(B) = hB.$$

Then, with $Q = q^L$, every $m_t = q B (1 + Q + \dots + Q^{t-1})$, $t \geq 1$, is a zero of S_χ .

Proof. With $s = 1$ and $T_q = -qh$, Theorem 4.1 reads $S_\chi(qk+r) = qS_\chi(k) + qkD(r) + T(r)$. Appending the digits of B one at a time, by induction, $S_\chi(Qk+B) = QS_\chi(k) + Qk \sum_i D(d_i) + S_\chi(B)$, so $S_\chi(k) = hk$ implies $S_\chi(Qk+B) = h(Qk+B)$; starting from $k = 0$, $k_t = B(1 + Q + \dots + Q^{t-1})$ has $S_\chi(k_t) = hk_t$. Appending the digit 0, $S_\chi(qk_t) = q(S_\chi(k_t) + k_t D(0)) = q(hk_t - hk_t) = 0$. ■

At $q = 59$, where $h = 3$, the block $B = (1, 22, 21, 0)_{59} = 283200$ has $\sum D = -2 + 3 + 2 - 3 = 0$ and $S_\chi(B) = 849600 = 3B$, so S_χ vanishes at $59B = 16\,708\,800$ and infinitely often after it. This is not the first zero: $77821 = (22, 21, 0)_{59}$ has $S_\chi(77821) = 3 \cdot 77821$, so the last step of the proof gives $S_\chi(4\,591\,439) = 0$ at $4\,591\,439 = 59 \cdot 77821$, but its digits have $\sum D = 2$ and do not close a loop. At $q = 347$, where $h = 5$, the block $B = 122\,338\,879\,485\,769\,202\,918\,256\,559\,803\,529$, of thirteen digits in base 347, has $\sum D = 0$ and $S_\chi(B) = 5B$, and does the same. Class number greater than one is therefore no obstruction to infinitely many zeros. Blocks are the simplest loops of the recursion, which carries a relation $S_\chi(k) = uk$ to another; other loops work as well. At $q = 131$, where $h = 5$, the digit 43 has $C(43) = 5$ and $T(43) = 86 = 2 \cdot 43$, so it preserves $S_\chi(k) = 2k$, and a final digit 27, with $C(27) = 3$ and $T(27) = 0$, sends $S_\chi(131k + 27)$ to $262k - 262k = 0$: every

$$27 + 43(131 + 131^2 + \dots + 131^t), \quad t \geq 0,$$

is a zero, its first three members being 27, 5660, 743583.

The mechanism of Corollary 4.3(b) does not need a quadratic character.

Corollary 4.5 Let q be an odd prime, χ an odd Dirichlet character modulo q of any order, extended completely multiplicatively by any root of unity $\chi(q)$, and $S_\chi(m) = \sum_{n \leq m} n\chi(n)$. Then $S_\chi(\frac{q-1}{2}) = 0$ if and only if $\chi(2) = 1$, and in that case $S_\chi(\frac{q^j-1}{2}) = 0$ for every $j \geq 1$. The characters for which this happens are the odd characters trivial on $\langle 2 \rangle$.

Proof. Theorem 4.1 and its proof hold verbatim for χ , with $s = \chi(q)$. By Theorem 4.2, $T((q-1)/2) = \frac{q}{2} B_{1,\chi}(\chi(2)^{-1} - 1)$ with $B_{1,\chi} \neq 0$, which gives the equivalence at $j = 1$. If $\chi(2) = 1$, the proof of Theorem 4.2 gives $C((q-1)/2) = (\chi(2)^{-1} - 2) B_{1,\chi} = -B_{1,\chi}$, and $T_q = qB_{1,\chi}$, so $(q-1)/2 \in R_q$. ■

So $\chi_{q,s} \in \mathcal{Z}_\infty$ for $q \equiv 1 \pmod{4}$ and for $q \equiv 7 \pmod{8}$, for both signs; and for $q \equiv 3 \pmod{8}$, where the half sum of Theorem 4.2 is $q h(-q)$ for $q > 3$ and not 0, for $q = 3$ and the primes of class number one with $s = 1$, wherever the recursion closes a loop, as at $q = 59, 131, 347$, and, with $s = -1$, always, at the lengths $(q^{2i} - 1)/2$ (Corollary 4.3(f)).

4.3 The same two sums in Conrey's sine series

For $q \equiv 3 \pmod{4}$, $q > 3$, the sums C and T are the ingredients of the sine series that Conrey studies,

$$f_q(x) = \sum_{n \geq 1} \left(\frac{n}{q}\right) \frac{\sin 2\pi nx}{n^2}.$$

His Theorem 4 [Con24], stated for squarefree $q \equiv 3 \pmod{8}$, uses only that the character is real, odd and primitive, and holds as it stands for prime $q \equiv 3 \pmod{4}$; combined with Dirichlet's class number formula, as in his Corollary 1, it reads, in the present notation and for $q > 3$ with $h = h(-q)$,

$$f_q(x) = \frac{2\pi^2}{q^{3/2}} F_q(qx), \quad F_q(y) = T(r) + y(h - C(r)), \quad r = \lfloor y \rfloor,$$

for $0 \leq x < 1$, that is $0 \leq y < q$, where $0 \leq r < q$ and $C(r), T(r)$ are defined; at $x = 1$ the value is that of the continuous extension, $F_q(q) = T_q + qh = 0$, and f_q is periodic of period 1. So F_q is continuous on $[0, q]$, linear on each cell $[r, r+1]$, $0 \leq r < q$, and vanishes identically on that cell exactly when $C(r) = h$ and $T(r) = 0$, that is, when $r \in R_q$. These are the intervals of the first alternative of [Con24, end of §6] recalled in Section 1, and the digit set of Corollary 4.3 is exactly the set of them.

Proof of Corollary C. The zeros $S_\chi((q^j - 1)/2) = 0$ are Corollary 4.3(b). By the same corollary $(q - 1)/2 \in R_q$, so F_q vanishes identically on the cell $[(q - 1)/2, (q + 1)/2]$, and f_q on $[\frac{q-1}{2q}, \frac{q+1}{2q}]$. ■

When R_q is larger, f_q vanishes on further cells: at $q = 103$ also on $[\frac{47}{103}, \frac{48}{103}]$ and its mirror. For $q \equiv 3 \pmod{8}$, Conrey's case, such intervals are excluded under a positivity hypothesis of the kind he proposes.

Proposition 4.6 *Let $q \equiv 3 \pmod{8}$, $q > 3$, and let $\chi_{q,-1}$ be the extension with value -1 at q . If $\sum_{n \geq 1} \chi_{q,-1}(n) \sin(2\pi nx)/n^2 \geq 0$ for $0 \leq x \leq 1/2$, then $R_q = \emptyset$.*

Proof. By the symmetry $C(q-1-r) = C(r)$, $T(q-1-r) = q(C(r)-h) - T(r)$ it suffices to exclude $r \leq (q-1)/2$. Suppose such an r lies in R_q and put $u = (rq+1)/q^2 \in [0, 1/2]$. Splitting n by its q -adic valuation, the series at u is $\sum_{j \geq 0} (-1)^j q^{-2j} f_q(q^j u)$. The term $j = 0$ vanishes, qu lying inside the cell of r ; the terms $j \geq 2$ vanish, their arguments being integers; and the term $j = 1$ is $-q^{-2} f_q(1/q) = -2\pi^2 h/q^{7/2}$, since $F_q(1) = h$. The series is negative at u . ■

The proposition asks for positivity of the series of $\chi_{q,-1}$, not of f_q itself: the term $j = 1$ is what makes a null cell strictly negative, whereas a non-strict positivity of f_q alone would allow such cells.

5 Legendre symbols at a fixed length

For a quadratic character χ write $T(m, \chi) = \sum_{c=1}^m c \chi(c)$. When χ has values ± 1 on $1, \dots, m$, this is a signed sum of the first m integers with multiplicative signs, and parity decides at which lengths it can vanish.

Proposition 5.1 Let $n = 2m + 1$ and let ε be any function on $1, \dots, m$ with values ± 1 . Then

$$\sum_{c=1}^m c \varepsilon(c) \equiv \frac{n^2 - 1}{8} \pmod{2}, \quad \text{so it can vanish only if } \left(\frac{2}{n}\right) = 1.$$

In particular, if χ is a primitive quadratic character of conductor f with $\gcd(f, m!) = 1$, then $T(m, \chi)$ does not vanish unless $\left(\frac{2}{n}\right) = 1$.

Proof. Changing one sign changes the sum by $2c$, so its parity is that of $\sum_{c \leq m} c = m(m+1)/2$, and $m(m+1)/2 = (n^2 - 1)/8$. The second supplementary law of quadratic reciprocity is $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$, so the sum is even exactly when that symbol is 1. ■

Both hypotheses are needed. A quadratic character vanishes at the integers it shares a factor with: at $f = 35, m = 6$ the values are $1, -1, 1, 1, 0, -1$ and $1 - 2 + 3 + 4 + 0 - 6 = 0$ while $\left(\frac{2}{13}\right) = -1$. And $\left(\frac{2}{n}\right)$ is the Jacobi symbol, so the condition reads $n \equiv \pm 1 \pmod{8}$, which is $m \equiv 0, 3 \pmod{4}$.

Call row m the set of odd primitive quadratic characters χ of prime conductor $f > 2m + 1$ with $T(m, \chi) = 0$. Each solution ε of $S_\varepsilon(m) = 0$ imposes a set of Legendre conditions on f , that is, a union of reduced residue classes modulo $8 \prod_{3 \leq p \leq m} p$; the row $m = 3$, for instance, comes from the single solution $1 + 2 - 3 = 0$, that is $(2 | f) = 1$ and $(3 | f) = -1$.

Corollary 5.2 Let $n = 2m + 1$. The row m is non-empty if and only if $\left(\frac{2}{n}\right) = 1$, and then it is infinite.

Proof. Necessity is Proposition 5.1. Conversely, $\left(\frac{2}{n}\right) = 1$ is $m \equiv 0, 3 \pmod{4}$; take ε from Theorem A. For a prime $f \equiv 3 \pmod{4}$, $f > n$, the value $(2 | f)$ is fixed by $f \pmod{8}$, and for an odd prime $p \leq m$ reciprocity gives $(p | f) = (-1)^{(p-1)/2} (f | p)$, fixed by $f \pmod{p}$ and taking both signs. By the Chinese remainder theorem and Dirichlet's theorem there are infinitely many primes $f \equiv 3 \pmod{4}$ with $(p | f) = \varepsilon(p)$ for every prime $p \leq m$; for them $\chi = (\cdot | f)$ agrees with ε on $1, \dots, m$, and $T(m, \chi) = S_\varepsilon(m) = 0$. ■

6 Two questions

Question 6.1 For $q \equiv 3 \pmod{8}$, is R_q empty for every q ? By Proposition 4.6 it is for each q at which the series of $\chi_{q,-1}$ is non-negative on $[0, 1/2]$, and a computation shows it is for every $q < 10^5$. When moreover $h(-q) > 1$ and $s = 1$, no item of Corollary 4.3 gives infinitely many zeros: up to 10^7 , $q = 59$ has one zero, $q = 179, 227, 347$ none, and at $q = 179$ and 227 there is none with at most 40 digits in base q ; yet at $q = 59, 131, 347$ a loop of the recursion gives infinitely many. Does every $\chi_{q,1}$ with $q \equiv 3 \pmod{8}$ close a loop, or are there q with no zero at all? For $s = -1$ the answer is yes for every q : S_χ vanishes at $(q^{2i} - 1)/2$ for every i (Corollary 4.3(f)).

The zero of $q = 59$ below 10^7 is 4591 439. The search by digits works as follows. For $s = 1$ and $D(r) = C(r) - h$, Theorem 4.1 gives, for $k = qj + r$ with $0 \leq r < q$, that $S_\chi(k) + ak = b$ holds if and only if $q | b - T(r) - ar$ and $S_\chi(j) + (a + D(r))j =$

$(b - T(r) - ar)/q$. Reading the digits of m from the least significant one and starting from the pair $(a, b) = (0, 0)$, m is a zero exactly when every step is integral and the pair reached when the digits are exhausted, where $j = 0$, has $b = 0$. The pairs reached after d digits form a finite set, computed exactly level by level; two words reaching the same pair have the same continuations, so it suffices to keep one word per pair, provided each word is tested as it is read, before pairs are merged: a word is a positive zero when its leading digit, the last one read, is nonzero and the pair it reaches has $b = 0$. Run to $d = 40$, this finds no zero at $q = 179$ and 227 (with 58 and 148 pairs at the last level), and it returns 4 591 439 at $q = 59$. The pairs are unbounded integers in general, so this is an exhaustive search over the words of at most 40 digits, not a finite automaton.

Question 6.2 The number $a(m)$ of sign patterns on the primes $\leq m$ with $S_\varepsilon(m) = 0$ begins 1, 1, 1, 1, 2, 2, 2, 4, 1, 9, 5, 14, ... on the admissible lengths; a related unweighted sequence is [OEIS]. A Gaussian prediction conditioned on the signs of the primes up to $\sqrt{m}$ matches $a(m)$ within 3% for every admissible $80 \leq m \leq 120$. What is the asymptotic behaviour of $a(m)$?

The prediction is the following. Every $n \leq m$ has at most one prime factor $p > \sqrt{m}$, to the first power, so, once the signs of the primes $p \leq \sqrt{m}$ are fixed, $S_\varepsilon(m) = R + \sum_{\sqrt{m} < p \leq m} A_p \varepsilon(p)$ with $A_p = p S_\varepsilon(\lfloor m/p \rfloor)$, where R and the A_p depend only on the fixed signs. With $V = \sum_p A_p^2$, the prediction is $2^{\pi(m)} \mathbb{E} [2(2\pi V)^{-1/2} e^{-R^2/2V}]$, the factor 2 being the spacing of the values of $S_\varepsilon(m)$ and the expectation being taken exactly over the $2^{\pi(\sqrt{m})}$ choices of the small signs. For the admissible $80 \leq m \leq 120$ the ratio of $a(m)$ to it lies between 0.972 and 1.015. This is a heuristic comparison, not a theorem.

Supplementary material. The supplementary material contains the programs, with their outputs, that generate and check the 104 block certificates and the 10000 explicit functions of Theorem A and the inequalities of its analytic range; the certificates (each q_0 and its pairs) and the functions (their signs at the primes) stored as files; an independent program that only reads these files and checks every function, every block and the inequalities with the factor 16 of Proposition 3.1 in exact integer arithmetic; the examples of Section 4, the computations quoted in Section 6, and the Lean 4 file mentioned in Section 4.

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Use of artificial intelligence. Claude (Anthropic) and ChatGPT (OpenAI) were used as assistants, under the author's direction, in preparing the manuscript and the verification programs. No statement rests on their output: the finite range of Theorem A rests on integer certificates, stored in the supplementary material and checked there by an independent program, the digit recursion and the propagation of its zeros are formalized in Lean, and every computation quoted is produced by a program in the supplementary material. The author is responsible for the content.

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